

Splunk SOC Dashboard Project

Security Monitoring & Threat Forensics Documentation

PROJECT EXECUTIVE SUMMARY

This project demonstrates the implementation of a functional **Security Operations Center (SOC)** environment using Splunk Enterprise. By ingesting raw Apache access logs, the system provides real-time visibility into malicious web activity, automated scanning signatures, and high-risk exploit attempts.

SIEM

Log Analysis

SPL

Threat Hunting

Cybersecurity

CORE VISUALIZATIONS

- **Global Attack Origin (Map):** Utilizes Geo-IP mapping to track geographic concentrations of malicious requests, identifying high-risk IP origins.
- **Attacker Tool Breakdown (Pie Chart):** Identifies signatures of automated vulnerability scanners (e.g., Nikto) versus standard user-agents.
- **Detailed Exploit Logs (Statistics Table):** Provides deep forensic data, capturing exact attack payloads for CVE-2014-6271 (Shellshock) and Cross-Site Scripting (XSS).

TECHNICAL IMPLEMENTATION (SPL)

The primary logic for the forensic dashboard table uses the following Splunk Processing Language (SPL) query:

```
index=* sourcetype=access_combined
| stats count, values(method) as methods, dc(status) as status_codes by
clientip, useragent
| rename count as "Total Requests", useragent as "Attacker Tool"
```

DETECTED VULNERABILITIES

- 1. Shellshock (CVE-2014-6271):** Captured exploit strings attempting command injection via HTTP headers: `() { :; }; echo...`
- 2. Cross-Site Scripting (XSS):** Identified script injection attempts within URL parameters and user-agent fields: `<script>alert(1)</script>`

ARCHITECTURE & TOOLS

- **Platform:** Splunk Enterprise 10.2.2
- **OS Environment:** Linux (Parrot Security OS)
- **Data Source:** Apache2 Access Logs (`/var/log/apache2/access.log`)
- **Methodology:** Field extraction, data normalization, and visualization dashboarding.